

Informe de experimento: react_nogate

Generado el 2026-09-18 22:45 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio react_nogate (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- System prompt: hash 7da5e2a37ec4 (40 filas)
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-18 21:49
- Costo total: 1.3020 USD (40 items; costo medio 0.03255 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=458416
- Solved global: 100.0 % (40/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método react_nogate (system_prompt_hash real de esta fila: 7da5e2a37ec4)

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
```

3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del **scoreboard_per_case**), en los 4 grupos de **metricas_pfagent_definicion**
utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta −toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_automously_rate (no la columna Solved de arriba, que puede solaparse con Escalated −una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100 %: dónde termina cada petición −resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Escalated	Wrong (silent)	\$
react_nogate	case1440	100.0 (40/40)	0	0.00	100.0 (40/40)	100.0 (40/40)	0	90.0 (36/40)	−	2.6	11460−	0.0 (0/40)	0.0 (0/40)	0.0 (0/40)	1.3020

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
react_nogate	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
react_nogate	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)	100.0 (10/10)

Tipos de error de formulación (conteo de items)

Method	ok
react_nogate	40

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5930 chars más}
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 3 ---
LLM: 1. Summary: I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers:
- Total load: 253.8458 MW
- Total generation: 266.5103 MW
- Total losses: 12.6645 MW
- Voltage range: 1.0100 to 1.0900 pu

3. Violations, if any:
- Voltage vi ...[409 chars más]
--- Respuesta final ---
1. Summary: I loaded IEEE case14 and disconnected the line between bus 10 and bus 11; the power flow converged successfully.

2. Key numbers:
- Total load: 253.8458 MW
- Total generation: 266.5103 MW
- Total losses: 12.6645 MW
- Voltage range: 1.0100 to 1.0900 pu

3. Violations, if any:
- Voltage violations only; no thermal violations were reported.
- Overvoltage buses: 1 (1.0600 pu), 6 (1.0700 pu), 7 (1.0573 pu), 8 (1.0900 pu), 11 (1.0636 pu), 12 (1.0536 pu)
- The disconnected line 10-11 now has 0.0 MW flow, as expected.

4. Suggested next step:
- If you want, I can check which branch is now most heavily loaded, run an N-1 contingency analysis, or suggest remedial actions to reduce
↳ the overvoltages.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5819;
↳ cost_usd=0.0180; wall=5.8 s
```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-multistep-030-s0, case14, difficulty=multistep, seed=537958268): Load case14, set the active load at bus 14 to 17.2 MW,
↳ disconnect the line between bus 1 and bus 5, reconnect the line between bus 1 and bus 5, and finally run an N-1 contingency analysis
↳ ranked by min voltage and report the 8 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 14, "p_mw": 17.2}) -> disconnect_line({"from_bus": 1,
↳ "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8, "criteria": "min_voltage"})
executed_calls: load_case({"case_name": "case14"}) -> set_active_load({"bus_id": 14, "p_mw": 17.2}) -> disconnect_line({"from_bus": 1,
↳ "to_bus": 5}) -> reconnect_line({"from_bus": 1, "to_bus": 5}) -> run_n1_contingency({"top_k": 8, "criteria": "min_voltage",
↳ "max_candidates": 0})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 271.9322468585065,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: set_active_load({"bus_id": 14, "p_mw": 17.2})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.452601937946843, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 6367 chars más}
gate: PASS (reasons=-, mismatch=6.8e-09 MW)
--- Ronda 3 ---
tool: disconnect_line({"from_bus": 1, "to_bus": 5})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -8.384501830561186, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 6071 chars más}
gate: PASS (reasons=-, mismatch=2.94e-08 MW)
--- Ronda 4 ---
tool: reconnect_line({"from_bus": 1, "to_bus": 5})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.452601937946843, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 6265 chars más}
gate: PASS (reasons=-, mismatch=6.8e-09 MW)
--- Ronda 5 ---
```

```

tool: run_n1_contingency({"top_k": 8, "criteria": "min_voltage", "max_candidates": 0})
output: {"plot_type": "n1_ranking", "figure_json":
  ↳ {"data": [{"cliponaxis": false, "hoverinfo": {"text": "\u003c\u003eOutage\u003c\u003e: 3-4",
  ↳ (line)\u003cbr\u003e\u003c\u003eConverged\u003c\u003e: True\u003cbr\u003e\u003c\u003eVoltage
  ↳ violations\u003c\u003e ...[19510 chars más]
  ↳ gate: n/a (no es un resultado de flujo)
--- Ronda 6 ---
LLM: 1. **Summary:** I loaded IEEE case14, set bus 14 active load to **17.2 MW**, disconnected and then reconnected line **1-5**, and
  ↳ finally ran an **N-1 contingency analysis** on the restored network ranked by **minimum voltage**, reporting the **8 worst outages**.

2. **Key numbers:**
- Total load: ...[1728 chars más]
--- Respuesta final ---
1. **Summary:** I loaded IEEE case14, set bus 14 active load to **17.2 MW**, disconnected and then reconnected line **1-5**, and finally
  ↳ ran an **N-1 contingency analysis** on the restored network ranked by **minimum voltage**, reporting the **8 worst outages**.

2. **Key numbers:**
- Total load: **275.1414 MW**
- Total generation: **290.6577 MW**
- Total losses: **15.5163 MW**
- Voltage range: **1.0100 to 1.0900 pu**
- Final base case after reconnecting line 1-5: **converged**

3. **Violations:**
- **Base case voltage violations:** 8 overvoltage violations at buses **1, 6, 7, 8, 9, 10, 11, 12**
- **Base case thermal violations:** none
- During the temporary outage of line **1-5**, the case still converged; after reconnection, the system returned to the above base-case
  ↳ stat ...[1228 chars más]
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=stale_state; error=-; tokens=39592;
  ↳ cost_usd=0.1079; wall=13.7 s

```

Ejemplo: primer ítem

```

request (case14-ambiguous-007-s0, case14, difficulty=ambiguous, seed=2073320062): Load the 14-bus test case (case14) and disconnect the
  ↳ line between bus 6 and bus 13 (1-based).
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 261.9028323717408,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 6, "to_bus": 13})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.104476346710827, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va ...[5772 chars más]
gate: PASS (reasons=-, mismatch=5.12e-08 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 6 and bus 13; the post-contingency power flow converged.

2. Key numbers
- Total load: 261.90 MW
- Total generation: 276.66 MW
- Total losses: 14.76 MW
- Voltage range: 0.9989 to 1.0900 pu

3. Violations, if ...[398 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14 and disconnected the line between bus 6 and bus 13; the post-contingency power flow converged.

2. Key numbers
- Total load: 261.90 MW
- Total generation: 276.66 MW
- Total losses: 14.76 MW
- Voltage range: 0.9989 to 1.0900 pu

3. Violations, if any
- Voltage violations only; no thermal violations were reported.
- Overvoltage buses:
- Bus 1: 1.0600 pu
- Bus 6: 1.0700 pu
- Bus 7: 1.0572 pu
- Bus 8: 1.0900 pu
- Bus 11: 1.0531 pu

4. Suggested next step
If you want, I can now check the updated network status in more detail, identify the most loaded branch, or run an N-1 contingency analysis
  ↳ for this modified 14-bus system.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=5793;
  ↳ cost_usd=0.0179; wall=5.7 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- Todos los métodos que llegan al solver (react_nogate) tienen $V_MAE < 1e-4$ p.u. en todos los casos (máximo 0 en react_nogate/case14).
- El gate verificó 52 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de **react_nogate**: 100.0 % (40/40).
- Formulación exacta agregada sobre métodos con tools: 100.0 % (40/40).
- Formulación por dificultad: mejor **plain** 100.0 % (10/10), peor **plain** 100.0 % (10/10).
- Ningún item reclamó éxito sobre un fallo (claimed_success_on_failure=0).
- 1 de 15 items con mutación de red reportaron estado obsoleto (stale_state).
- Trazabilidad de números (Faith.) media con tools: 98.9 %.
- Costo total 1.3020 USD; el método más caro fue **react_nogate** con 1.3020 USD (100 %).
- Métodos con tools: 2.65 tool calls medias por item (máximo 5).

6. Archivos

Rutas absolutas bajo **react_nogate_report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/report.json    [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard.rescored.json
scoreboard_per_case.rescored.json:
↳ /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/p
↳ ower-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_load_split/gpt-5.4/react_nogate/traces
```